Real-Time Dense Map Matching

Naive hidden Markov models: delay versus accuracy

G Harish reddy | PES1UG24AM366

SHARANABASAVA H | PES1UG24AM399

UE24CS352A Machine Learning

Problem and motivation

GPS noise can place a vehicle on the wrong road.

Estimate the hidden directed road arc for each reading.

Waiting N readings provides future evidence but delays output.

Avoid shortest-path searches; study the measured trade-off.

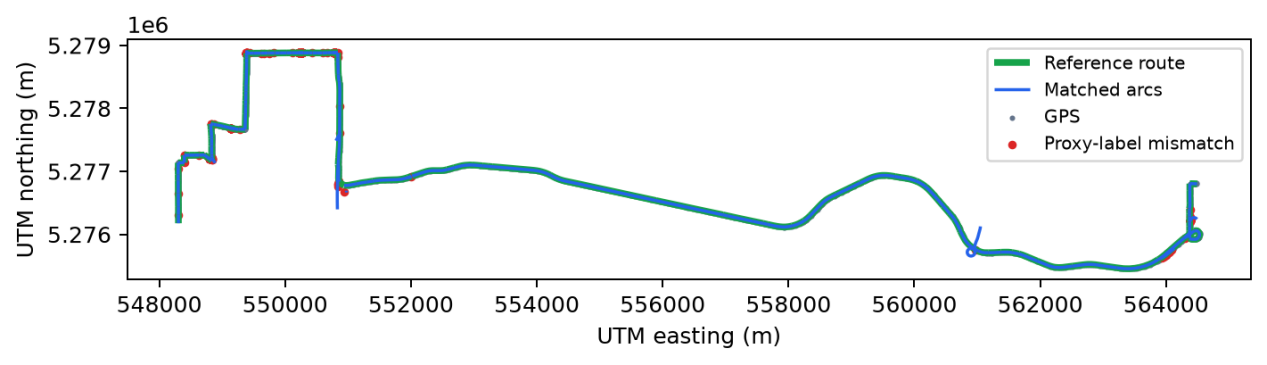
Seattle benchmark

7,531 readings; 158,167 road edges; 576 reference-route edges.

Nominal 1 Hz with four gaps; coordinates projected to metres.

60% train / 10% validation / 30% held-out test.

Point labels derived from route alignment: evaluation proxies.



Pipeline

GPS + roads -> nearby directed arc candidates

Distance, heading, speed features -> emissions

History and displacement features -> transitions

Log-domain sliding-window Viterbi -> irrevocable predictions

HMM and features

States: all directed arcs within 50 m; sigma = 10 m.

Emission: Gaussian distance, heading dot product, speed constraint.

Transition: backtrack prevention and midpoint discrepancy.

Positive weighted potentials; no connectivity guarantee.

Window T and buffer N

At reading t, solve over max(0, t-T+1) through t.

Clamp overlapping committed states at their actual times.

Commit prediction for t-N; never revise it later.

Flush final N points at EOF; report online-only error separately.

Learning feature weights

Balanced L1 LinearSVC; candidates/pairs labelled from reference.

10 expanding folds, 15-reading gap; C in {0.1, 1, 10}.

Selected C: emission 10, transition 10.

Positive coefficients normalized; intercept discarded.

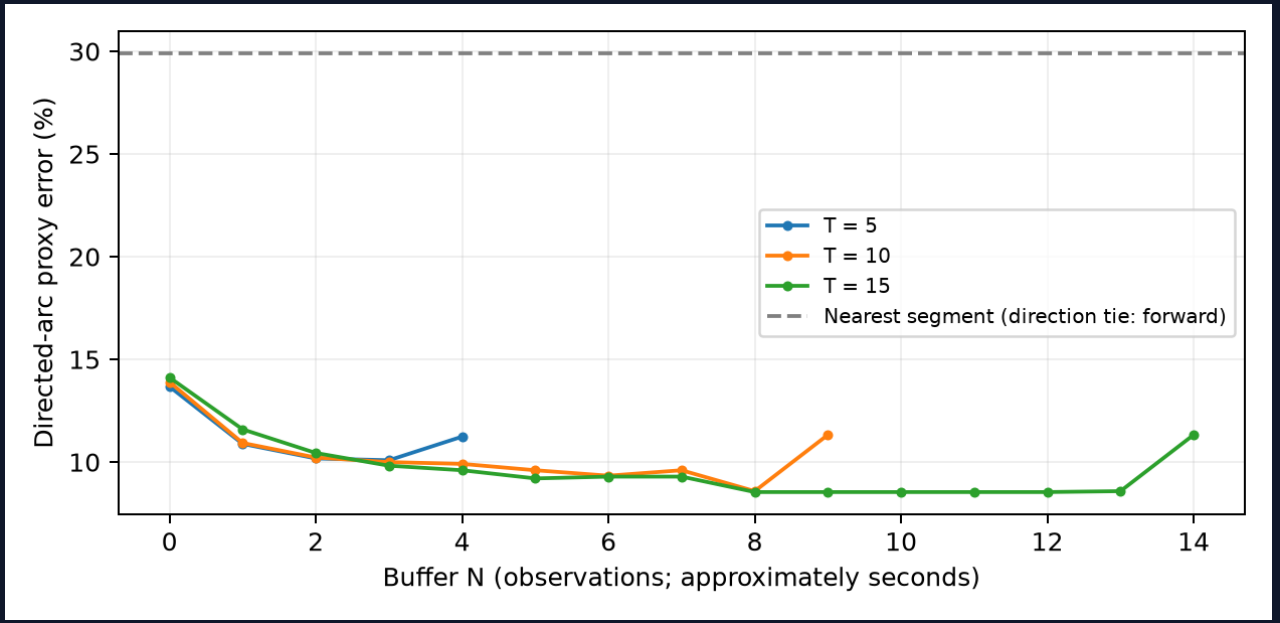
Delay versus accuracy

Validation selected T=10, N=7.

Test directed error: 9.60%.

Test edge error: 4.60%.

Minimum is not always at T/2; curves are not uniformly convex.



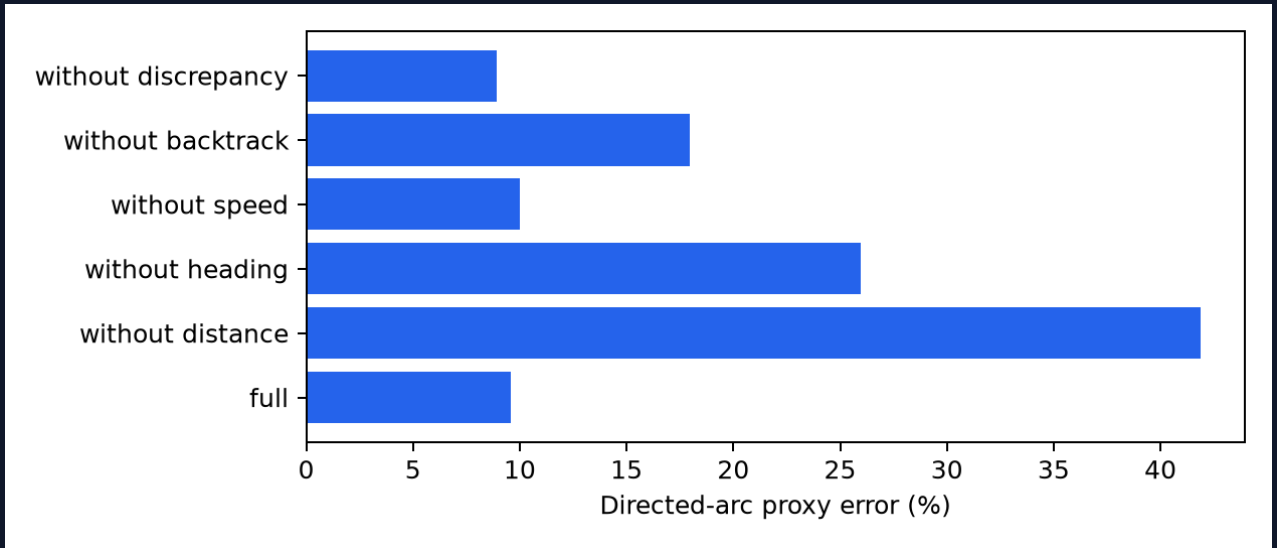
Baseline and ablations

Nearest-road edge error: 8.01%.

Full model edge error: 4.60%.

Removing heading or distance sharply worsens error.

Removing discrepancy improves directed error to 8.94%.



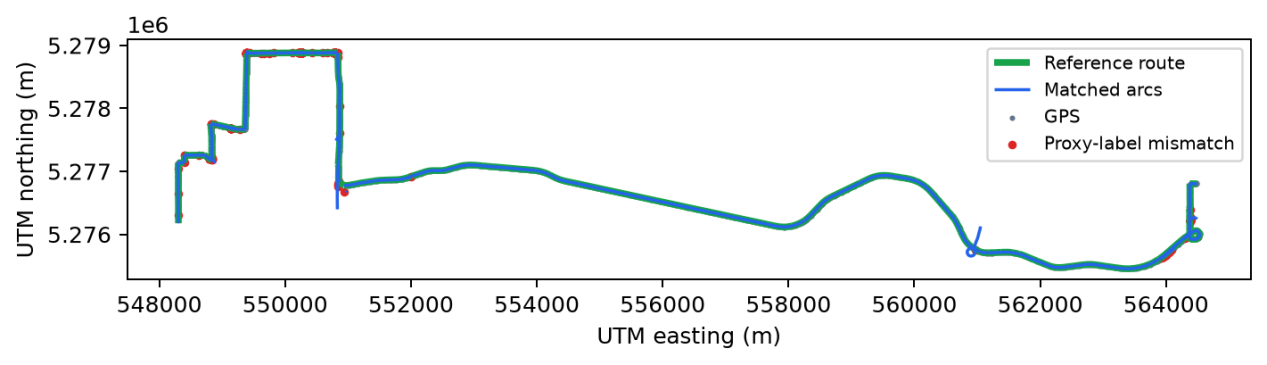
Live demonstration

python run_demo.py --T 10 --N 7 --slice 5300 5600

Open results/demo_map.html: completely offline replay.

Compare N=0 and N=7 using the same slice.

Static PNG is the backup; runtime excludes map rendering.



Conclusions and limitations

Buffering and heading help on this single benchmark.

Derived labels and one trip limit generalization.

Training/inference history differ; no network connectivity check.

Next: independent labels, multiple trips, filtering, turn features.

Code, contribution and Q&A

src/: loader, candidates, features, Viterbi, training, evaluation.

Tests verify causal commitments and future-evidence effects.

Both members should review the full pipeline and run the demo.

Record actual individual contributions; private GitHub publishing pending.